



Deep Learning-Based Wheat Rust Detection and Severity Estimation

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Problem Statement

Plant rust significantly reduce agricultural productivity.

Farmers often rely on manual inspection, which is slow and inaccurate.

Incorrect diagnosis leads to improper fungicide use and financial loss.

There is a need for a fast, reliable, and automated detection system.

Introduction

- Plant rust appear as visible patterns on leaves (spots, discoloration, texture changes).
- Deep learning, especially CNNs, can detect these patterns effectively.
- Image-based disease detection provides a scalable and accessible solution.
- This project focuses on classifying plant leaves as healthy or diseased.



Relevant Apps / Research

- Existing apps like **Plantix** and **LeafSnap** provide plant/disease identification.
- Research shows CNN-based models achieve high accuracy in image classification tasks. (Marcos, A. P., Rodovalho, N. L. S., & Backes, A. R. (2019, September). Coffee leaf rust detection using convolutional neural network. In *2019 XV Workshop de Visão Computacional (WVC)* (pp. 38-42). IEEE.)
- Transfer learning models (ResNet, MobileNet) are widely used in similar studies.
- These solutions inspire automated and real-time disease detection systems.

Dataset

Wheat Plant Diseases Dataset is used for training.

Contains thousands of labeled images of healthy and diseased leaves.

Includes multiple plant species and disease categories.

Well-structured and commonly used in deep learning research.



Proposed Method

01

Collect and preprocess leaf images (resize, normalize).

02

Use a pretrained CNN model (e.g., ResNet/MobileNet).

03

Apply transfer learning to improve accuracy.

04

Train model to classify images into rust categories.

05

If asked give severity level of rust and how to deal with it.

Expectations / End Goal

-  Develop an accurate plant disease detection model.
-  Achieve high classification accuracy .
-  Build a simple interface for user interaction.
-  Help farmers detect diseases early and take preventive action.
-  Reduce crop loss and improve agricultural efficiency.